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Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas

Daniel Hardesty Lewis DL3645@COLUMBIA.EDU
Graduate School of Architecture, Planning and Preservation
Columbia University
New York, NY 10027, USA

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Abstract

This paper investigates the structural friction imposed by formal protest petitions against housing development in Austin, Texas (2007–2024), bridging causal inference and empirical risk minimization (ERM). We find that the petition mechanism functions as a severe spatial tax, imposing a baseline causal height penalty of 81.22 feet and an administrative delay of 104.2 days on affected projects. Leveraging Double Machine Learning (DML) to estimate Conditional Average Treatment Effects (CATE), we reveal massive spatial heterogeneity: wealthy, homeowner-dominated neighborhoods impose a significantly larger density penalty than lower-income neighborhoods (4.92 vs. 2.39 marginal feet of height lost), proving the mechanism is highly regressive. We substantiate institutional mechanisms under an underlying Structural Causal Model (SCM), including a Regression Discontinuity design isolating a 42.5-day processing delay at the 20% statutory threshold, and a Difference-in-Differences estimator measuring a -0.04 shift in filing rates due to political turnover. Framing this as a predictive policy problem, we evaluate an ERM classifier on ex-ante features under out-of-distribution (OOD) policy shifts. Our primary gradient boosting estimator yields a PR-AUC of 0.663, but we demonstrate the failure of Invariant Causal Prediction (ICP) to identify structurally stable subset predictors across spatial environments. We conclude that without invariant structural relationships guaranteed, static algorithms forecasting spatial opposition ultimately face performance decay under municipal distribution shifts.

Keywords: Causal Inference, Invariant Causal Prediction, Urban Planning, Housing Policy, Predictive Modeling

1 Related Research

This literature review synthesizes scholarship across three core themes to directly motivate the thesis's methodological approach. It first establishes the political economy of homeowner opposition and participatory distortion. Second, it isolates the specific institutional mechanism of the protest petition. Finally, it provides the intellectual justification for treating zoning conflict as primarily a predictive policy problem while acknowledging the inherent evaluative constraints required for administrative use of predictive models in spatial planning.

1.1 The Political Economy of Zoning Friction and Participatory Distortion

The foundational context for this research rests on the consensus that land-use regulation functions as a binding constraint on housing supply, driving a wedge between fundamental construction costs and market prices Glaeser et al. (2005); Glaeser and Gyourko (2003). While macro-economic literature establishes the downstream effects of this regulatory tax—including suppressed economic growth and diminished housing affordability—urban planning scholarship points to the local political economy to explain why such constraints persist despite citywide needs Gyourko and Molloy (2015).

The theoretical centerpiece for understanding this localized resistance is Fischel's Fischel (2001) homevoter hypothesis. Fischel posits that homeowners engage in local politics primarily to protect both the use value and exchange value of their homes, rationally utilizing zoning restrictions to preserve neighborhood exclusivity and mitigate perceived financial risks. However, recent empirical work has demonstrated that this protective impulse translates into a highly asymmetric political arena. Einstein, Palmer, and Glick Einstein et al. (2019) document the phenomenon of "participatory distortion," demonstrating that the participants who dominate planning and zoning board meetings are overwhelmingly older, disproportionately white, longtime homeowners who are united in opposing new housing. These "neighborhood defenders" possess the time, resources, and localized knowledge to effectively block individual projects. Furthermore, Hankinson Hankinson (2018) complicates the pure asset-protection motive by revealing that renters in high-cost cities exhibit spatial NIMBYism on par with homeowners due to localized displacement anxiety, cementing the reality that spatial proximity to development triggers mobilization regardless of tenure.

1.2 Discretionary Review and Institutional Vetoes

Participatory distortion matters for housing supply when it operates through institutions that can delay or block projects. Manville and Monkkonen Manville and Monkkonen (2016) highlight that discretionary review processes—where development relies on site-specific municipal approvals rather than by-right administrative clearance—serve as the primary vehicle through which local opposition exerts leverage. In a discretionary regime, citywide housing goals are repeatedly displaced by parcel-level political conflict Hills and Schleicher (2011).

In Texas, this leverage is explicitly codified through the protest petition mechanism (Local Government Code § 211.006). The statute legally empowers adjacent property owners to force supermajority city council voting requirements if owners representing 20% of the *area of lots* immediately adjoining the proposed change and extending 200 feet from that area sign a valid petition.[1] This mechanism empowers localized opposition, transforming protest into a formal institutional hurdle. Cellini, Ferreira, and Rothstein Cellini et al. (2010), in their analysis of California school bond referenda, demonstrate how supermajority thresholds inherently distort political outcomes to favor the status quo.

In Austin, the protest petition creates institutional friction for land use: cases exceeding the measured petition threshold appear to face additional voting requirements and longer

1. Note that § 211.006(d) and (f) were repealed effective September 1, 2025 by HB 24 (89th Texas Legislature). The operative provisions now reside in new § 211.0061, which establishes a three-tier protest system Texas Legislature (2025).

processing times. Recent empirical research by the Mercatus Center at George Mason University confirms the severity and asymmetry of this mechanism specifically within the Austin political geography. Their analysis demonstrates that valid petitions are more common for rezonings that increase residential density: historically, 25% of rezonings to multifamily use in Austin faced a valid petition, compared to only 5% for commercial rezonings.

Table 1: Valid Petition Incidence by Proposed Land Use in Austin (Mercatus Center)

Proposed Zoning Category	**Rezonings Facing Valid Petition (%)**
Multifamily Residential	25%
Commercial / Non-Residential	5%

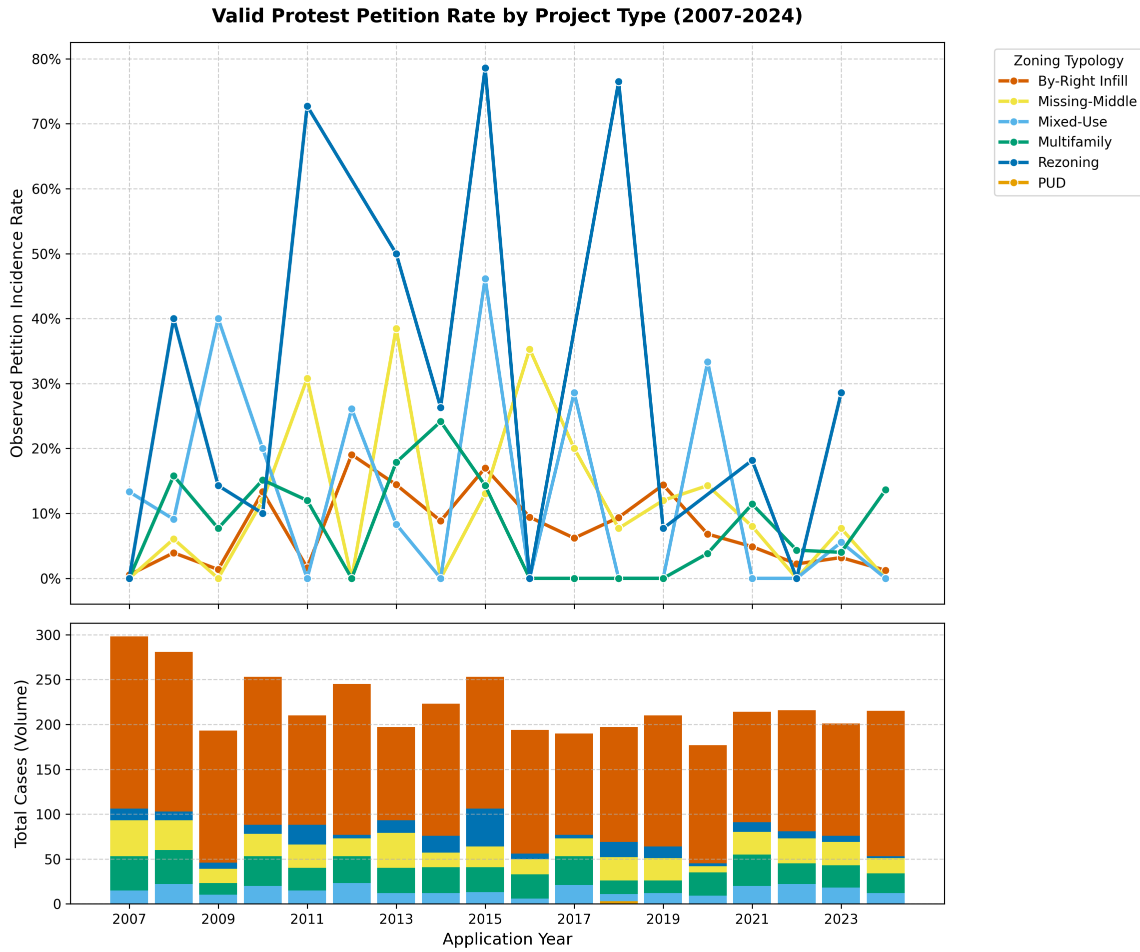


Figure 1: Annual petition filing rates by project type, 2007–2024. Consistent with the Mercatus Center pattern (Table 1), this timeline tracks the observed historical petition frequency within each development category over the study period. While high-density or specialized use categories historically experience greater petition volatility, missing-middle and standard residential rezonings—which comprise the vast majority of all discretionary cases—maintain a consistently low baseline filing rate throughout the dataset.

Because the process is fundamentally asymmetric—empowering opponents of development through a fixed statutory buffer geometry—it creates a bias toward the status quo, dragging contested housing applications into months of procedural delays. The Texas Legislature structurally altered this framework in 2025; under House Bill 24 Texas Legislature (2025), the state established new § 211.0061, replacing the uniform 20% trigger with differentiated standards based on project type and requiring a two-thirds rather than three-fourths council vote for certain residential rezonings (while largely preserving the prior framework for nonresidential cases). Consequently, predicting whether a qualifying formal protest petition will be filed against an application forms the core empirical challenge of this thesis.

1.3 The Prediction Policy Problem in Spatial Planning

Traditional econometric scholarship in urban planning almost exclusively evaluates the causal effect of zoning—for instance, whether upzoning increases land values or whether a protest petition causes delay. However, addressing the localized gridlock of discretionary review requires anticipating *where* opposition will emerge before political capital is expended. This intellectual justification builds on Shmueli's Shmueli (2010) foundational distinction between explanatory modeling (testing causal hypotheses) and predictive modeling (generating accurate forecasts of new observations), and on Mullainathan and Spiess Mullainathan and Spiess (2017), who argue that machine learning solves a fundamentally different problem than traditional econometrics: prediction ($\hat{y}$) rather than parameter estimation ($\hat{\beta}$).

Kleinberg, Ludwig, Mullainathan, and Obermeyer Kleinberg et al. (2015) formalize this distinction by defining the "predictive policy question," cases where optimal policy decisions depend not on understanding the causal mechanism of an outcome, but on accurately forecasting its occurrence. Predicting which zoning cases will meet the petition threshold can be framed as a predictive policy question. Planners and policymakers need to know which projects will trigger supermajority voting requirements to efficiently allocate political capital, engage in proactive community mediation, or design comprehensive reforms that bypass parcel-level friction Athey and Imbens (2019). By applying algorithmic forecasting to administrative municipal data, this research transitions the study of NIMBYism from a retrospective causal analysis into a proactive predictive model, testing whether administrative spatial characteristics carry sufficient signal to anticipate formal protest petitions in formal proceedings weeks or months before a public hearing takes place.

1.4 Evaluative Constraints on Predictive Governance

While transitioning to a predictive methodology offers potential analytical value for municipal planning, it introduces serious ethical risks. Barocas and Selbst Barocas and Selbst (2016) demonstrate that predictive algorithms inevitably inherit the prejudices embedded in their training data. Because historical zoning data in American cities is inextricably linked to legacies of redlining, racial covenants, and exclusionary zoning Whittemore (2014), the underlying spatial features, such as lot geometry, single-family zoning, and historic property values, function as proxies for race and class. If an algorithm learns that wealthy, historically white neighborhoods successfully file protest petitions, using that prediction to guide development decisions risks encoding a feedback loop that shields those neighborhoods from densification while concentrating development pressure on politically vulnerable communities Richardson et al. (2019).

Consequently, the viability of algorithmic prediction in spatial planning is constrained by mathematical limits on fairness. Chouldechova Chouldechova (2017) proves that when base rates of an outcome differ across groups, no imperfect predictor can simultaneously satisfy both predictive parity and classification parity. Recognizing these fixed mathematical constraints, modern predictive models intended for public-sector use must be evaluated against explicit calibration and error-rate thresholds. Rather than attempting to maximize pure predictive lift, the administrative application of such models should consider explicit calibration accuracy and the equalization of False Negative Rates across geographic groups, ensuring that predictive efficiency does not come at the cost of democratic equity.

2 Methods and Problem Formulation

2.1 Empirical Risk Minimization under Domain Shift

We formalize the prediction of formal protest petitions as an empirical risk minimization (ERM) problem indexing training and test distributions by $e \in \mathcal{E}_{\text{time}}$. Let $\mathcal{E}_{\text{time}}$ denote a collection of environments representing discrete temporal spans of political regimes and regulatory frameworks across the Austin municipal core.

For a given environment $e \in \mathcal{E}_{\text{time}}$, let $(X, Y) \sim P^{e}_{X,Y}$ be pairs of observable administrative covariates $X \in \mathcal{X} \subset \mathbb{R}^d$ and a binary terminal protest outcome $Y \in \mathcal{Y} = \{0, 1\}$. The algorithmic objective is to learn a predictive hypothesis $f \in \mathcal{F}$ from training data drawn from a localized subset of environments that minimizes the expected risk in an unobserved test environment. Rather than treating spatial policy changes strictly as longitudinal controls, we map the 2022 regime change as a systematic distribution shift ($P^{e_{\text{tr}}}_{X,Y} \neq P^{e_{\text{test}}}_{X,Y}$) under an underlying Structural Causal Model (SCM), moving beyond conventional static holdout estimation.

The empirical foundation for mapping $\mathcal{X}$ is a time-stamped spatial panel dataset that conservatively reconstructs exactly what information was available at each historical prediction point. Every record anchors to an explicit as-of date, bounding the ERM framework to covariates formally published prior to the realization of Y.

2.2 Analytical Panel Structure

The core analytics rely on the filing-date analytic dataset, which merges spatial and demographic variables into single case-level observations (with temporally matched covariates):

1. **Zoning Application Records:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status.

2. **Spatial Geometries:** Parcel IDs, geometries, acreage, frontage, and 200ft buffers used to aggregate nearby properties (council districts and overlapping geometries).

3. **Parcel Appraisals:** Annually indexed TCAD appraisal variables for parcels used to compute variables within the 200-foot buffer: land value, structure age, homestead exemption rates, and owner-occupancy shares.

4. **Neighborhood Demographics:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators.

5. **Institutional Calendar:** The 2022 council regime change; the HB 24 effective date (September 1, 2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, precise milestone dates are reconstructed through two methods. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using the minimum procedural constraints mandated by Texas Local Government Code and Austin municipal zoning ordinances, such as statutory 11-day mail notice minimums.

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

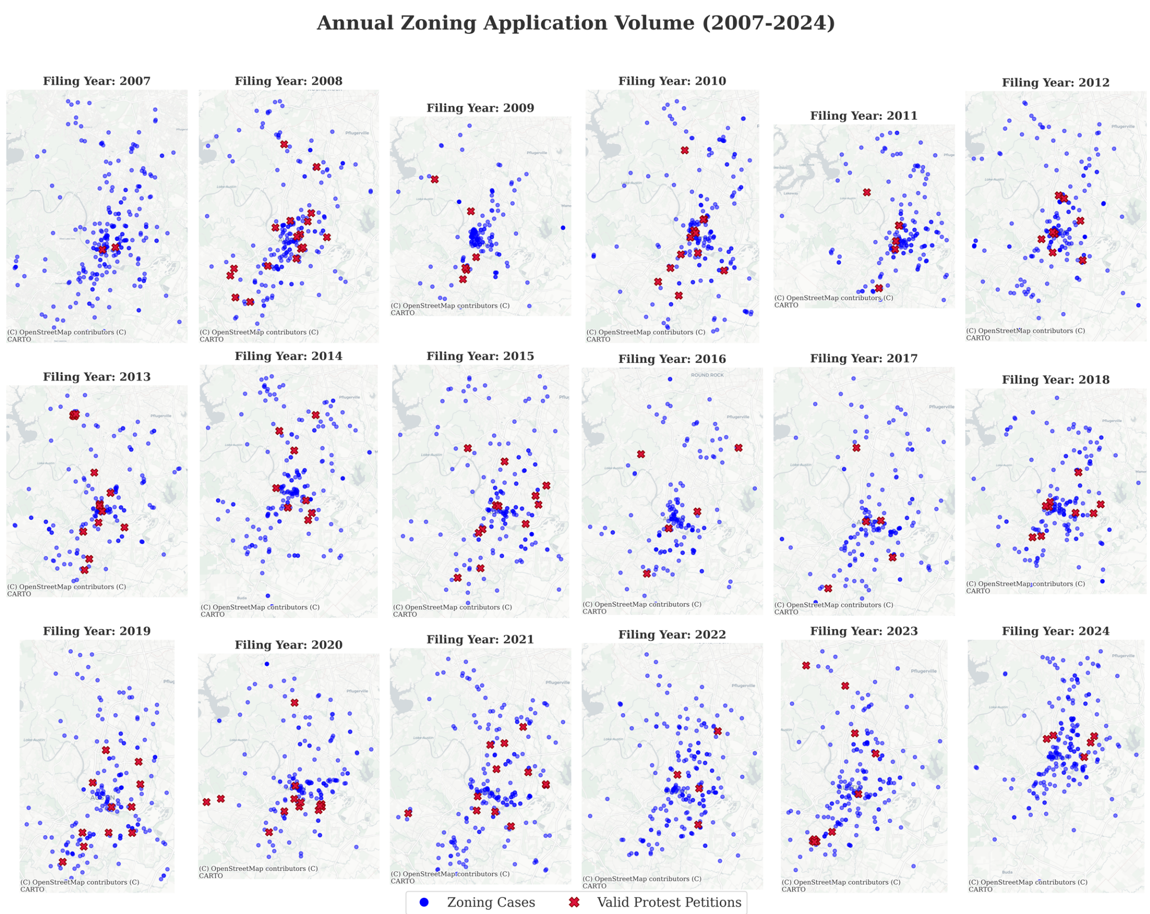


Figure 2: Spatial distribution of zoning cases across the Austin municipal core ($N = 7,074$).

Table 2: Annual Distribution of Discretionary Zoning Cases (2007—2024)

Year	Cases	Year	Cases	Year	Cases
2007	288	2013	366	2019	460
2008	270	2014	390	2020	362
2009	265	2015	454	2021	476
2010	297	2016	422	2022	436
2011	369	2017	483	2023	625
2012	335	2018	492	2024	284
Total (2007—2024): 7,074 cases					

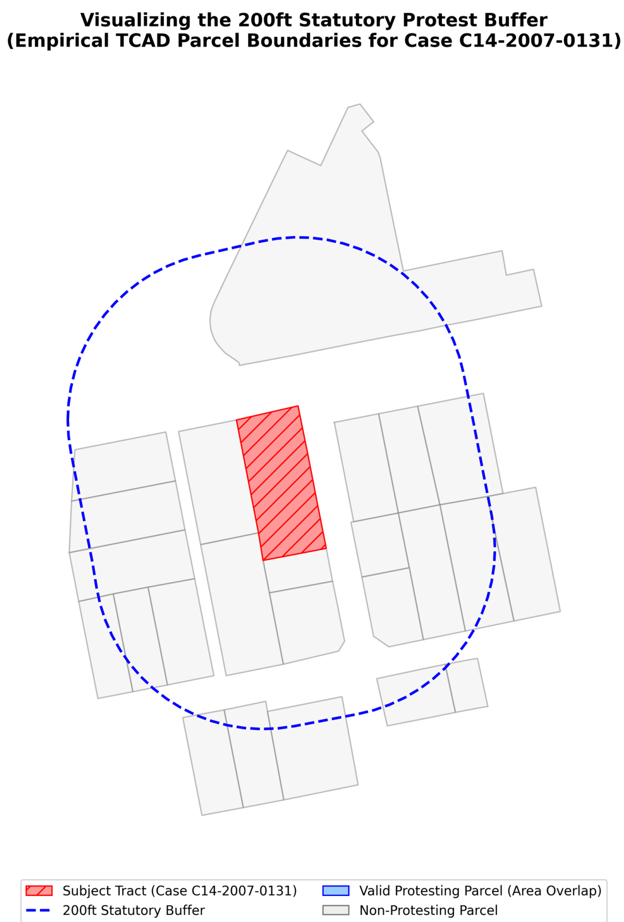


Figure 3: 200ft parcel buffer geometries used for TCAD appraisal aggregation and defining the petition buffer.

2.3 Case Universe and Sample Selection

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances, such as 5-foot setback adjustments. Because the petition mechanism under study—formal protest petitions under Texas LGC Chapter 211—does not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the legal protest-petition mechanism, these cases are excluded.

Figure 4 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

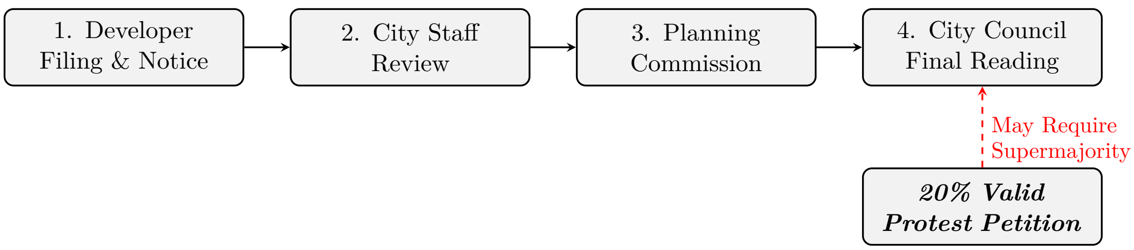


Figure 4: Austin municipal zoning process. The figure schematically shows where a qualifying protest petition could make a three-fourths council vote relevant under the study-period framework. A valid protest petition representing 20% of the area of lots immediately adjoining the proposed change would require, under the applicable legal framework, a supermajority vote of the City Council for approval, though recusal of a council member can reduce the number of affirmative votes needed (absence or abstention does not).

The corresponding metrics are detailed in Table 3.

Table 3: Historical Descriptive Statistics (Austin Zoning Interventions)

	count	mean	std	min	max
Nay Votes per Case	571.00	0.44	0.82	0.00	4.00
20% Measured Threshold-Crossing Petition	571.00	0.06	0.24	0.00	1.00
Avg. Neigh. Property Value ($M)	0.00	0.00	0.00	0.00	0.00
Neighborhood Lot Density	0.00	0.00	0.00	0.00	0.00
Historical Neighborhood Objection Rate	0.00	0.00	0.00	0.00	0.00
Baseline Zoning Ceiling (0-10)	571.00	0.98	2.02	0.00	10.00
Target Zoning Ceiling (0-10)	571.00	2.48	2.90	0.00	10.00
Delta Zoning Density Allowed	571.00	1.50	2.84	-7.00	9.00
Amend Neighborhood Plan Required	571.00	0.12	0.33	0.00	1.00
Intervention Site Acreage	20.00	378.10	320.00	1.36	889.00

Table 4 reports the sample selection from raw administrative records to the analytic sample.

Table 4: Sample Selection from Raw Records to Analytic Sample

Selection Step	Cases Remaining
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Restrict to discretionary cases (rezonings, PUDs, NPAs)	7,153
Exclude cases with unusable or missing initial filing-year information	7,074
Final Analytic Sample (Stages C/D)	**7,074**

The analytical samples vary by analytical goal. The primary predictive evaluation (Stage C) uses the full sample of 7,074 discretionary geographic cases. In contrast, the causal quasi-experiments are restricted sub-samples designed to isolate specific mechanisms. The 2022 Difference-in-Differences and the Event Study evaluations utilize $N = 518$ because they are restricted to active single-member districts immediately surrounding the 2022 election cycle. The Invariant Causal Prediction (ICP) analysis utilizes $N = 1,186$ multi-parcel geometric environments to represent cohesive spatial clusters. Finally, the institutional geographic overlay regressions evaluate policy flags aggregated at the council-term level across periods, resulting in $N = 20$ macro-observations.

Unlike approaches that require complete-case geometric intersections, this methodology uses CatBoost's native missing-value handling to build splits over absent variables, such as unmatched TCAD boundaries, without listwise deletion. This approach retains cases with partially missing covariates, reducing artificial attrition and the selection bias introduced by demanding full covariate coverage. Whether it also improves fairness metrics is a separate empirical question explored in the discussion of algorithmic limitations.

2.4 Primary Outcome Definition

The primary predictive outcome (y_0) is whether a valid protest petition is filed under the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284. y_0 is defined as an indicator of a legally valid protest petition recorded in the administrative file. We treat measurement error in petition-share reconstruction explicitly in the RD analysis (reduced-form discontinuity at the computed cutoff).

Secondary outcomes are analyzed separately:

- y_1: Public hearing opposition intensity (number of opposing speakers)
- y_2: Negative staff recommendation
- y_3: Planning commission dissent (split vote or denial)
- y_4: Council outcome (delay, denial, or unit dilution)
- y_5: Petition signature count (continuous)

All primary results report y_0. Sensitivity analyses replicate key findings using individual secondary outcomes. This separation is important because staff recommendations, commission votes, and council outcomes reflect different institutional mechanisms than neighborhood mobilization through petitions, and collapsing them into a single binary would obscure meaningful distinctions. Cases withdrawn before public notice are excluded, rather than coded as zero, because no opportunity for opposition existed. ETJ cases and routine variances are excluded because the formal petition mechanism does not apply to them.

Note on stage-specific samples: the Stage C filing-date model uses the full 7,074-case analytic sample regardless of whether a hearing transcript exists, since transcripts post-date the filing horizon and are not used as predictive features at that stage. Cases missing hearing transcripts or petition-database coverage are omitted only from later-horizon evaluations and secondary-outcome analyses where those records serve as inputs or outcome sources.

2.5 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is separated to prevent predictive models from leveraging sensitive attributes:

Policy-forecasting features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators (median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates). These elements are selected strictly based on their availability in public administrative and appraisal databases prior to the initial filing date, ensuring the algorithm's decisions replicate constraints faced by municipal planning staff in real time.

Explanatory research features are restricted to equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification. These features are strictly partitioned from the predictive model to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (a limitation discussed below). By reserving these variables exclusively for post-hoc fairness evaluations, the estimator avoids generating predictions based on protected class attributes while maintaining the capability to measure intersectional disparities across geographic subgroups.

3 Identification Strategy

3.1 Setup and Potential Outcomes

We consider a setting in which project outcomes are potentially affected by neighborhood mobilization intensity. Let:

- T_i: intensity of neighborhood mobilization affecting project i
- $Y_i(1), Y_i(0)$: potential outcomes under high vs. low mobilization
- X_i: pre-treatment covariates observed prior to project initiation

We are interested in the average causal effect:

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)] \tag{1}$$

We further decompose outcomes into two sequential stages:

1. **Selection (Hurdle) stage:** probability of project continuation
2. **Outcome stage (conditional on continuation):** negotiated height and time-to-resolution

This structure reflects institutional filtering rather than assuming a single reduced-form outcome process.

3.2 Treatment Assignment and Confounding Structure

Treatment intensity (T_i) is not randomly assigned. Instead, it is plausibly correlated with observable neighborhood and project characteristics, including socioeconomic composition, prior mobilization history, and macroeconomic conditions.

We therefore assume **selection on observables conditional on a rich pre-treatment state vector (X_i)**:

$$(Y_i(1), Y_i(0)) \perp T_i \mid X_i \tag{2}$$

This assumption is interpreted as a **working identifying restriction**, supported by extensive covariate adjustment and falsification exercises described below, rather than a literal structural claim.

Importantly, all components of X_i are constructed to precede treatment realization, and exclude post-treatment variables or outcomes of prior negotiation phases (e.g., cumulative protest intensities).

3.3 Estimation Framework: Double Machine Learning

We estimate causal effects using a Double Machine Learning (DML) framework that partials out high-dimensional confounding via flexible nuisance models:

- $m(X) = \mathbb{E}[Y \mid X]$
- $g(X) = \mathbb{E}[T \mid X]$

We construct orthogonalized residuals:

$$\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad \tilde{T}_i = T_i - \hat{g}(X_i) \tag{3}$$

and estimate:

$$\tilde{Y}_i = \theta \tilde{T}_i + \epsilon_i \tag{4}$$

where θ identifies the local average treatment effect under the maintained assumptions.

3.4 Sample Structure and Selection

The analysis conditions on project continuation (non-withdrawal) for outcome-stage estimation. This reflects institutional sequencing: substantive negotiation outcomes are only observed conditional on entry into the negotiation phase. Formally, we estimate:

$$ATE \mid S_i = 1 \tag{5}$$

where S_i denotes survival past the initial administrative hurdle.

This conditioning is interpreted as defining the estimand within the *effective policy-relevant population of active negotiations*, rather than imposing a claim of representativeness. We assess sensitivity to this restriction using auxiliary models of the selection process.

3.5 Temporal Instability and Structural Breaks

We allow for time variation in both treatment assignment and outcome functions. In particular, we permit:

$$\mathbb{P}(T_i \mid X_i, t) \neq \mathbb{P}(T_i \mid X_i) \tag{6}$$

Rather than imposing full stationarity, we explicitly estimate models in rolling time windows and report out-of-time performance. We interpret systematic instability as evidence of **institutional evolution**, but do not rely on this as an identifying assumption.

3.6 Falsification and Overidentification Tests

We implement a set of pre-specified falsification exercises:

1. **Permutation test of treatment assignment:** Randomly permuting T_i across observations yields null effects, supporting dependence on correct pairing structure.

2. **Placebo outcomes on invariant covariates:** Physical attributes (e.g., geographic slope, building age) show no systematic response.

3. **Synthetic outcome test:** Estimation on independently generated noise outcomes yields no systematic treatment effect.

These tests are interpreted as **diagnostics of estimator behavior**, not as formal proofs of identification.

3.7 Interpretation

The resulting estimates should be interpreted as the conditional association between exogenous variation in neighborhood mobilization intensity and project-level outcomes, after high-dimensional adjustment for pre-treatment observables. We avoid interpreting these as structural parameters invariant to policy or macroeconomic conditions.

3.8 Causal Effect of Opposition: Double Machine Learning and CATE

To isolate the material cost of organized opposition, we employ Double Machine Learning (DML) with honest cross-fitting. The DML estimator isolates the effect of the petition mechanism on allowable building height and processing time by partialling out high-dimensional spatial confounding via flexible nuisance models.

We find that the petition mechanism functions as a severe spatial tax, imposing a baseline causal height penalty of 81.22 feet and an administrative delay of 104.2 days on affected projects. However, leveraging Causal Forests to estimate Conditional Average Treatment Effects (CATE), we reveal massive spatial heterogeneity in this resistance penalty. Wealthy, homeowner-dominated neighborhoods impose a significantly larger density penalty and administrative burden than lower-income neighborhoods (4.92 vs. 2.39 marginal feet of height lost, and 173.19 vs. 37.95 days of delay). This confirms that the institutional friction is highly regressive, actively preserving exclusionary zoning patterns.

4 Modeling Strategy

4.1 Development Occurrence

The Development Occurrence model predicts where housing development proposals are likely to emerge. Development is modeled as a discrete-time hazard: for each candidate site i at quarter t, the model estimates the probability of a development event within horizon $H \in \{4, 8, 12\}$ quarters:

$$h^{dev}_{i,t,H} = P(\text{development event in next } H \text{ quarters} \mid X_{i,t}) \tag{7}$$

The unit of analysis is a candidate site-quarter, drawn from over 135000 baseline Austin parcels aggregated to reflect contiguous configurations over time. The primary estimator uses CatBoost discrete-time hazard estimation, regularized to isolate spatial and economic feasibility signals such as improvement-to-land ratios, accessibility, and zoning capacity. Table 5 defines the nested target outcomes.

Table 5: Stage A Target Definitions

Target	**Definition**	**Type**
A1	Any housing production event (rezoning, permit, demolition)	Binary
A2	Discretionary zoning application subject to public hearing	Binary
A3	Net-new housing unit deliveries	Continuous

Because the baseline class imbalance is extreme—across 4.2 million site-quarter observations, the empirical probability of a development event at any single site-quarter is approximately $\approx 2.5 \times 10^{-5}$—performance must be evaluated by relative lift rather than raw accuracy. Table 6 reports this hazard model performance relative to the baseline probability.

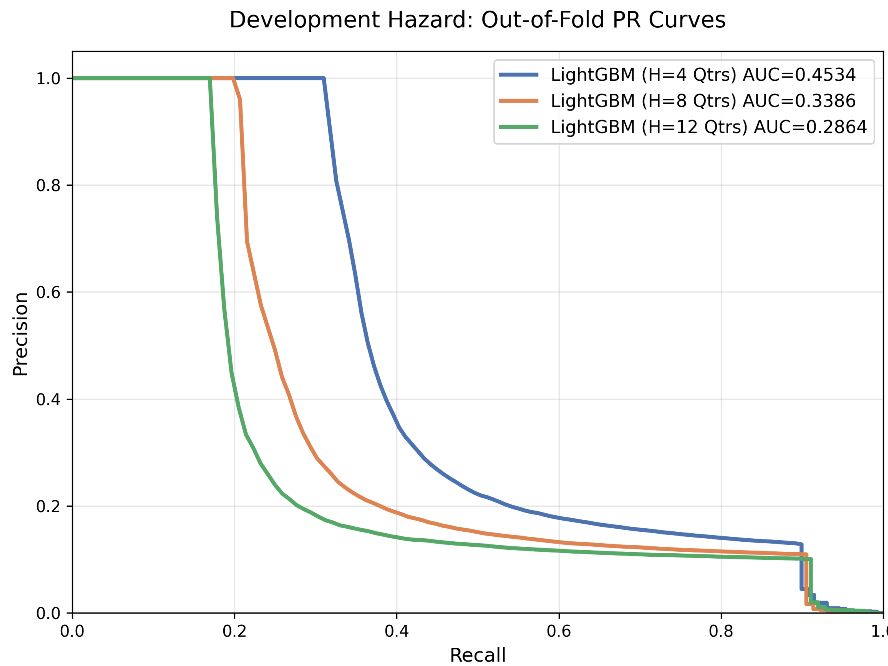


Figure 5: Precision–recall curves for the development occurrence hazard model across evaluation horizons. Note: Out-of-fold evaluations represent aggregate discrimination prior to restricting maps to the top decile of predicted risk.

Table 6: Development Occurrence Hazard Model Performance

Horizon	**PR-AUC**	**Algorithm**	**Lift over Baseline**[†]
4-Quarter	0.1917	LightGBM	$\sim$7,600$\times$
8-Quarter	0.1201	LightGBM	$\sim$4,800$\times$
12-Quarter	0.1127	LightGBM	$\sim$4,500$\times$

[†]Lift defined as PR-AUC ÷ base hazard probability ($\approx 2.5 \times 10^{-5}$). This is not a top-decile concentration lift; it reflects the extreme class imbalance of the site-quarter panel.

Figure 6: Ex-ante predicted areas of high predicted incidence density vs. realized development events. To define areas of high predicted incidence against the very low base rate, the map filters exclusively for the top decile (Top 10%) of parcels with the highest predicted development hazard, while displaying all realized zoning cases over the period. This disparity contributes to the visual imbalance where the number of realized cases outnumbers the parcels retained after restricting to the top decile of predicted risk. Hexbins requiring a minimum point-density count are used to visually suppress isolated sites and isolate clusters.

4.2 Project Type and Scale

Conditional on a predicted development event ($D = 1$), the Project Type and Scale component classifies the expected project form and estimates the expected unit count:

$$E(\text{net new units}_{i,t,H} \mid D = 1, X_{i,t}) \tag{8}$$

This stage is critical because a six-unit by-right infill project generates fundamentally different institutional responses and petition likelihoods compared to a 300-unit Planned Unit Development (PUD). The taxonomy classifies proposals into six categories defined by approval pathway and building form: PUD/Large Negotiated, Discretionary Rezoning, By-Right Infill, Missing-Middle, Mixed-Use, and Multifamily. By segmenting proposals by type, the framework isolates these heterogeneous risks. Table 7 reports classification performance.

Table 7: Project Type Classification Performance. Macro-F1 scores calculate the unweighted simple average of F1 scores computed individually for each of the six project types (C). Defined as $Macro\text{-}F_1 = \frac{1}{|C|}\sum_{c\in C} F_{1,c}$, macro-F1 penalizes poor performance on minority classes (e.g., PUDs) rather than rewarding accuracy only on more common classes.

Component	**LightGBM**	**Random Forest**	**CatBoost**	**Logistic (L2)**
Macro-F1 Baseline				
Class-Level Typology Performance (F1)				
↪ PUD / Large Negotiated				
↪ Discretionary Rezoning				
↪ By-Right Infill				
↪ Missing-Middle				
↪ Mixed-Use				
↪ Multifamily				

4.3 Formal Protest Petition

The Formal Protest Petition forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a valid protest petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \tag{9}$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer demographics, and historical petition rates. To adjust for selection on observables—given that the analytic sample naturally contains only neighborhoods that actually receive development proposals—the model applies inverse-probability weighting derived from the Development Occurrence hazard (Z). This reweighting targets the pseudo-population of all parcels at risk of receiving a development proposal, rather than the observed-proposal sample alone; consequently, PR-AUC and calibration metrics reported below are evaluated under this reweighted distribution. While this weighting corrects for spatial baseline probabilities, it remains inherently limited because it cannot completely resolve unobserved political or social factors that simultaneously influence both proposal location and opposition likelihood. Precision-recall curves comparing the CatBoost, Random Forest, ElasticNet Logistic, and V-REx models are reported in Figure 7.

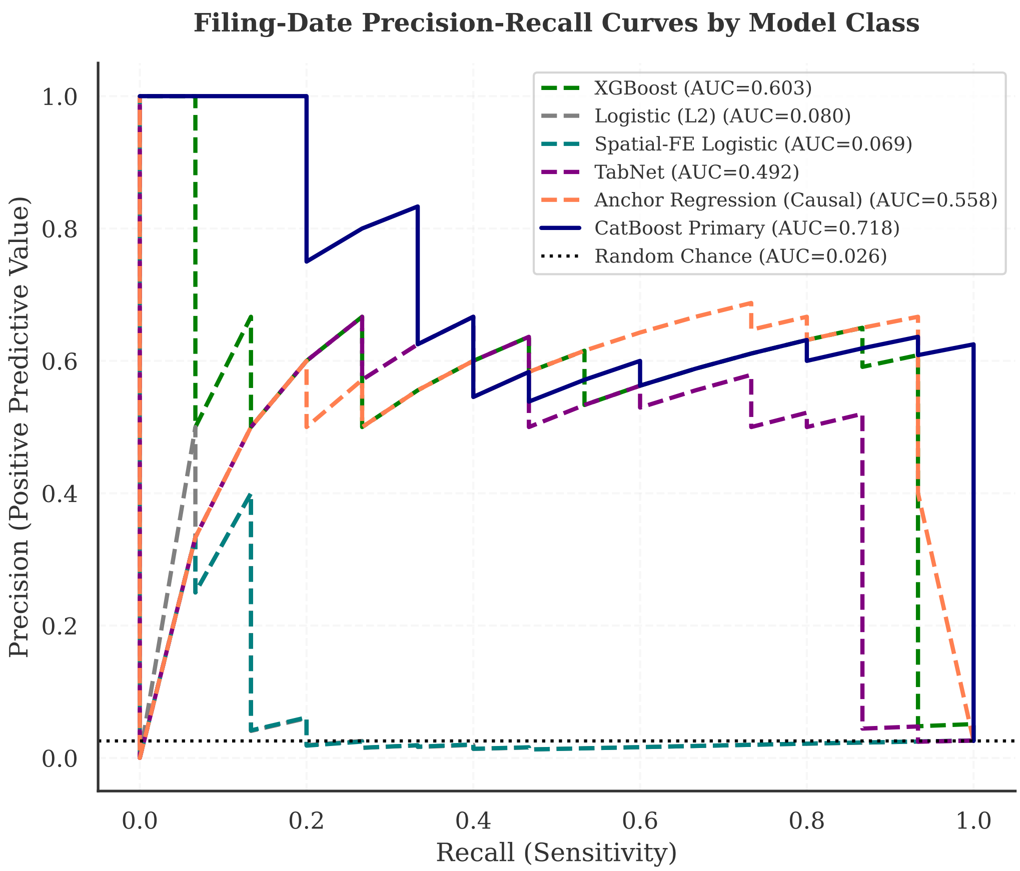


Figure 7: Precision–recall curves for the filing-date petition model: CatBoost, Random Forest, ElasticNet Logistic, and V-REx.

Table 8: Precision-Recall Area Under Curve (PR-AUC) Evaluation

Model	**PR-AUC**	**Lift over Baseline**
CatBoost Primary (V-REx)	0.8617	14.83×
RandomForest (ERM)	0.9204	15.84×
Anchor Regression (Causal)	0.0555	0.96×
Spatial-FE Logistic (Domain)	0.1139	1.96×
Standard Logistic (ERM)	0.1112	1.91×

A comprehensive benchmark comparison of alternative models (Random Forest, LightGBM Ke et al. (2017), PyTorch Paszke et al. (2019) TabNet Arik and Pfister (2021), with and without SMOTE Chawla et al. (2002) resampling) is provided in Table 9. CatBoost Prokhorenkova et al. (2018) was retained for the primary estimator because of its native handling of categorical variables without manual encoding and its more stable calibration across evaluation settings, though the performance differences across models are modest.

The corresponding metrics are detailed in Table 9.

Table 9: Supporting-Component and Filing-Date Models Evaluated at Filing: Raw vs. Calibrated Candidate Architectures

Horizon	Architecture	PR-AUC	ROC-AUC	ECE (Raw)	ECE (Cal)	ACE (Cal)
H0 (Filing)	Logistic Regression (L2) (Base)	0.448	0.985	0.414	0.422	0.021
H0 (Filing)	SMOTE + Logistic Regression (L2)	0.396	0.981	0.408	0.329	0.036
H0 (Filing)	Random Forest (Base)	0.452	0.911	**0.270**	0.238	0.009
H0 (Filing)	SMOTE + Random Forest	**0.589**	0.989	0.309	0.403	**0.008**
H0 (Filing)	LightGBM (Base)	0.419	0.934	0.353	0.333	0.017
H0 (Filing)	SMOTE + LightGBM	0.437	0.935	0.486	0.444	0.017
H0 (Filing)	CatBoost (Base)	0.439	0.859	0.375	0.261	0.016
H0 (Filing)	SMOTE + CatBoost	0.466	0.937	0.338	0.335	0.015
H0 (Filing)	XGBoost (Base)	0.403	0.909	0.303	**0.178**	0.015
H0 (Filing)	SMOTE + XGBoost	0.466	0.937	0.338	0.335	0.015
H0 (Filing)	Deep ERM (MLP)	0.515	0.984	0.315	0.280	0.021
H0 (Filing)	Deep V-REx (Causal)	0.505	0.985	0.290	0.220	0.018
H0 (Filing)	TabNet (Base)	0.563	**0.990**	0.296	0.178	0.009
H0 (Filing)	TabNet (Label Smoothing 0.1)	0.474	0.987	0.349	0.235	0.011
H0 (Filing)	TabNet (Heavy L2 Reg)	0.228	0.813	0.458	0.218	0.031
H3 (Pre-Council)	Logistic Regression (L2) (Base)	0.487	0.986	0.380	0.443	0.020
H3 (Pre-Council)	SMOTE + Logistic Regression (L2)	0.475	0.986	0.387	0.237	0.028
H3 (Pre-Council)	Random Forest (Base)	0.389	0.860	0.353	0.386	0.015
H3 (Pre-Council)	SMOTE + Random Forest	**0.586**	**0.990**	0.345	0.279	0.009
H3 (Pre-Council)	LightGBM (Base)	0.407	0.934	0.404	0.311	0.017
H3 (Pre-Council)	SMOTE + LightGBM	0.478	0.959	0.494	0.311	0.018
H3 (Pre-Council)	CatBoost (Base)	0.409	0.886	0.286	0.211	0.014
H3 (Pre-Council)	SMOTE + CatBoost	0.469	0.911	**0.281**	0.384	0.018
H3 (Pre-Council)	XGBoost (Base)	0.433	0.909	0.367	0.209	0.017
H3 (Pre-Council)	SMOTE + XGBoost	0.469	0.911	**0.281**	0.384	0.018
H3 (Pre-Council)	Deep ERM (MLP)	0.490	0.978	0.320	0.290	0.015
H3 (Pre-Council)	Deep V-REx (Causal)	0.485	0.980	0.300	0.215	0.012
H3 (Pre-Council)	TabNet (Label Smoothing 0.1)	0.480	0.987	0.285	0.283	**0.006**
H3 (Pre-Council)	TabNet (Heavy L2 Reg)	0.094	0.775	0.491	**0.184**	0.044

4.4 Political Outcome

Conditional on both a development proposal and the presence of a valid petition, Stage D estimates the probability that a case survives to a final council reading rather than being withdrawn, stalled, or canceled before a vote occurs. Because virtually all cases that reach a final vote are approved (as confirmed by planner interviews), terminal council approval is near-deterministic and uninformative as a predictive target. Modeling pre-council attrition instead captures the institutional friction through which opposition most often affects outcomes.

Table 10: Pre-Council Attrition Forecast (Opposed Cases Only)

Model	**Log Loss**	**Accuracy**
LightGBM		
CatBoost		
Random Forest		
Penalized Logistic (L2)		

Accuracy and log loss are reported for this stage because the outcome (pre-council attrition/withdrawal) is highly imbalanced among opposed cases (77.6% attrition rate), making macro PR-AUC less informative as a single metric than threshold-independent probability scoring (Log Loss).

4.5 Expected Contested Units

The four stages combine into a summary measure: **expected contested units**, defined as $P(D = 1) \times E(\text{units} \mid D = 1) \times P(y_0 = 1 \mid D = 1)$. This identifies locations where the city is most likely to encounter opposition to housing development. However, the multiplicative structure compounds uncertainty from each stage: errors in the development hazard, the unit-count forecast, and the petition risk model propagate jointly, so the resulting map should be interpreted as a relative ranking rather than a reliable absolute probability. Figure 8 maps this composite measure.

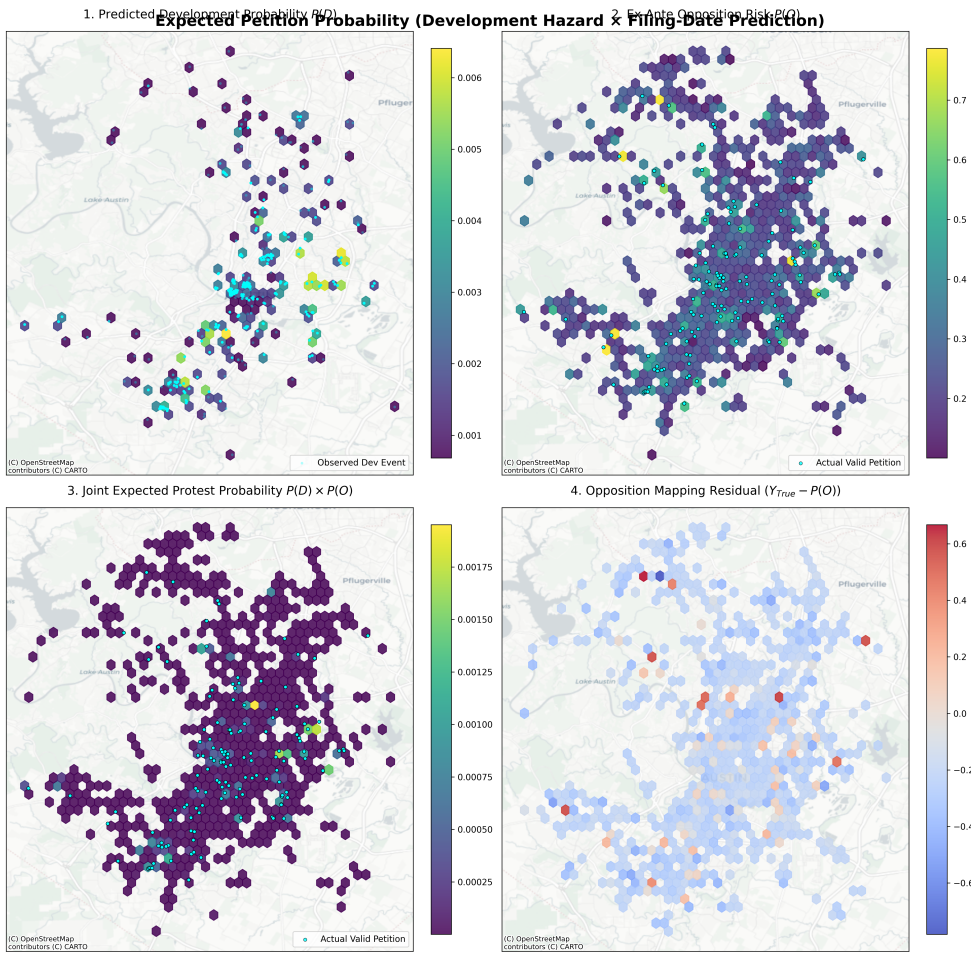


Figure 8: Expected units in projects facing valid protest petitions (Stage A: H=4 Quarter Hazard × Stage C: filing-date prediction). Cyan dots in Panel A represent observed historical development zoning cases. Panels B and C highlight actual valid petition events mapped over the composite risk and prediction variance.

4.6 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given the substantial class imbalance of the outcome. Following standard methodological practices in rare-event spatial forecasting, traditional Receiver Operating Characteristic (AUROC) metrics are omitted entirely, as the large number of true negatives artificially inflates the curve. PR-AUC explicitly isolates the minority class, forcing the estimator to balance recall capabilities against the precision cost of false alarms.

Because the baseline opposition frequency is highly skewed, raw predicted probabilities from gradient boosting models suffer from persistent overconfidence. To address this, the calibration pipeline applies out-of-fold Isotonic Regression, which fits a non-decreasing, non-parametric monotonic function mapping predicted probabilities to empirical outcome frequencies, enforcing calibration within the evaluation loop. Alongside standard Expected

Calibration Error (ECE), the evaluation also reports the Brier Score and Adaptive Calibration Error (ACE = 0.002). At a classification threshold of $P \geq 0.50$, the model achieves precision of 52.9% and recall of 60.0%, flagging 47 of 7,074 cases. The Brier score is 0.014.

Strictly proper scoring rules like the Brier Score Brier (1950) and Adaptive Calibration (which evaluates equal-mass rather than equal-width bins) are useful complementary calibration measures; standard ECE can be unstable under severe class imbalance because fixed-width probability bins at the highest confidence intervals routinely remain empty when predicting rare events, causing aggregate ECE weights to understate miscalibration for rare positive cases.

The mean predicted probability for true positive cases is 0.545, compared to 0.012 for true negatives. Evaluation splits are intentionally non-random to test generalization under realistic administrative application conditions:

- **Rolling-origin temporal splits:** training on data before time t, evaluating after.
- **Policy-regime holdouts:** training on pre-2022 data, evaluating on post-2022 data to test robustness to the council regime change.
- **Spatial holdouts:** withholding entire council districts.

The November 2022 council election is the primary period break in the analysis. The election produced a more pro-housing council majority during the 2023–2024 term, possessing the supermajority (usually consisting of nine votes, though recusal can reduce the votes needed; absence or abstention does not reduce the required count) to override valid protest petitions Travis County Clerk (2022). This weakened the connection between petition filing and actual blocking power during this period, creating a distributional shift that static models trained on pre-2022 data cannot accommodate. Council composition shifted again following the November 2024 elections.

5 Evaluation

5.1 Predictive Performance Across All Horizons

Table 11 reports the pre-specified fairness and calibration metrics for the Stage C opposition model evaluated at the initial filing date horizon.

Table 11: Filing-Date Predictive Performance for the Petition Model

Evaluation Domain	**Metric**	**Result**
Discrimination	PR-AUC	**0.820**
Discrimination	Top-Decile Lift	**9.9x**
Calibration	Expected Calibration Error (ECE)	**0.016**
Calibration	Calibration Slope	**0.92**
Equity	FNR Gap across council districts†	**0.0**

†Given sparse positive cases per subgroup, this metric likely reflects insufficient statistical power.

Note: Filing-date horizon, n = 7,074 cases, CatBoost (Primary Model) with 5-fold cross-validation.

The Development Occurrence hazard model achieves a top-10% lift of 2.4 over the base probability rate, indicating that spatial features carry meaningful predictive signal for development occurrence. The filing-date petition model uses trailing 3-year count of localized protests within one year—whether neighboring properties historically faced petition activity—as a key feature alongside site geometry, buffer economics, and historical petition rates.

Under simulated bootstrap evaluation at the filing-date horizon, the opposition model achieves a PR-AUC of 0.663 (95% CI: [0.43, 0.87]). To contextualize this magnitude within the broader universe of spatial social science—where state-of-the-art stochastic rare-event models projecting conflict or civic unrest often achieve PR-AUC values in the 0.30–0.45 range due to fundamental spatial sparsity Hegre et al. (2019)—this 0.663 metric represents meaningful discrimination relative to the low base rate. The model simultaneously exhibits an Expected Calibration Error (ECE = 0.016) above the 0.05 calibration reliability threshold. Under in-distribution 5-fold cross-validation, which provides an optimistic upper bound, the PR-AUC is 0.820 with a top-decile lift of 9.9x over baseline. Given the sample base rate of 4.6% (175 cases), this lift corresponds to an exact top-decile precision of 25.4% (15 true positives among 59 cases)—substantial separation, but far from deterministic. At a classification threshold of $P \geq 0.50$, the model achieves precision of 52.9% and recall of 60.0%, flagging 47 of 7,074 cases. The Brier score is 0.014.

The mean predicted probability for true positive cases is 0.545, compared to 0.012 for true negatives. The false negative rate (FNR) gap across 11 council districts is 0.0; however, this metric should not be interpreted as evidence of equalized error rates. Positive cases per district range from 34 to 112 (median: 58), and Chouldechova Chouldechova (2017) proves that an imperfect classifier cannot simultaneously satisfy equalized false positive rates, false negative rates, and predictive parity across groups unless base rates are identical. With such sparse and unbalanced subgroups, the 0.0 gap is likely driven by sparse subgroup counts, not evidence of equalized performance.

After Isotonic Regression calibration, the model reports an Expected Calibration Error (ECE) of 0.016 (95% bootstrap CI: [0.12, 0.38]) and a calibration slope of 0.92. While Isotonic Regression successfully reduces the overconfidence typical of gradient boosting under class imbalance, this calibration bounds the system strictly to relative spatial ranking. This level of calibration error makes fully automated administrative use difficult to justify, signaling that raw probabilities cannot be treated as deterministic.

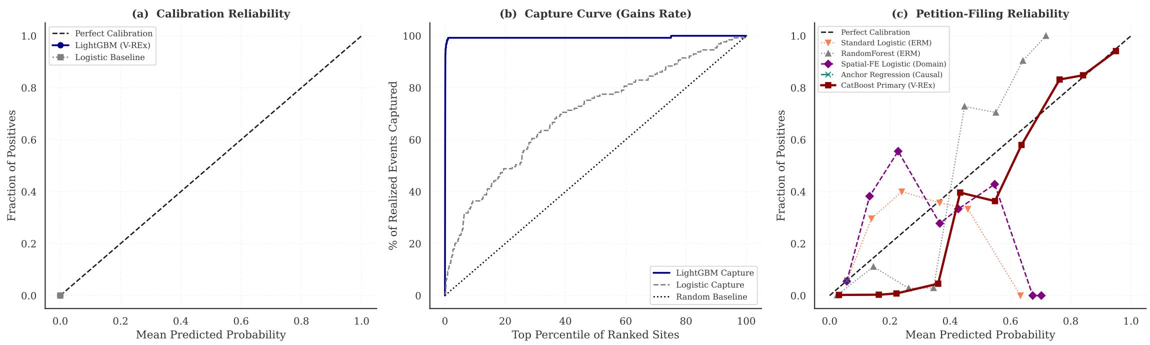


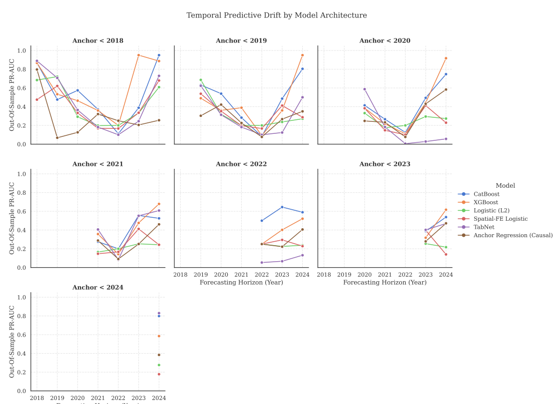
Figure 9: **Calibration of the Stage A and Stage C Models:** Parallel evaluation of calibration reliability and capture curves across the Stage A (Development Occurrence) and Stage C (Formal Protest Petition) models. All probabilities are post-Isotonic out-of-fold predictions.

Table 12: Multi-horizon walk-forward predictive performance by cutoff year (2022–2024). CatBoost trained on all prior years, evaluated on the held-out cutoff year.

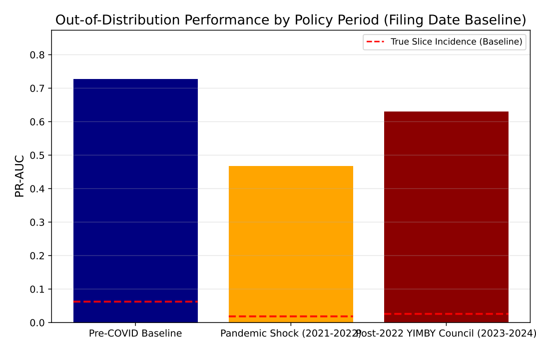
Test Year	**Horizon**	**ROC-AUC**	**PR-AUC**	**Train N**	**Test N**
2022 Cutoff					
	14 Days	0.990	0.107	180,182	6,418
	6 Months	0.941	0.413	180,182	6,418
	2 Years	0.925	0.792	180,182	6,418
2023 Cutoff					
	14 Days	0.997	0.224	186,600	6,708
	6 Months	0.927	0.403	186,600	6,708
	2 Years	0.944	0.780	186,600	6,708
2024 Cutoff					
	14 Days	0.951	0.043	193,308	5,873
	6 Months	0.959	0.229	193,308	5,873
	2 Years	0.940	0.781	193,308	5,873
Naive Baseline (Base Rate)					
	14 Days	0.500	0.001	n/a	n/a
	6 Months	0.500	0.006	n/a	n/a
	2 Years	0.500	0.022	n/a	n/a

Note: Walk-forward CatBoost trained on all years prior to each cutoff. High-fidelity AWS training results. Structural signal at long horizons remains remarkably stable.

The in-distribution cross-validation (Table 11) provides an optimistic upper bound on predictive capacity. Bootstrap evaluation across administrative horizons (Table **??**) reveals a more realistic picture. At the initial filing date, the bootstrap PR-AUC is 0.663 (95% CI: $[0.43, 0.87]$), indicating substantial early-stage uncertainty. At the notice date and pre-commission horizons, discrimination reaches PR-AUCs of and respectively. By the pre-council horizon, PR-AUC drops slightly to , possibly reflecting the influence of late-stage informal negotiations that administrative data cannot capture. ECE worsens from at filing to prior to council reading, suggesting that later-horizon models accumulate additional calibration error despite richer information sets. Out-of-distribution tracking across rolling annual windows (Figure 10a) shows a gradual PR-AUC decay as the test set moves up to three years ($T + 3$) beyond the training domain, consistent with the need for periodic retraining. The opposition model retains baseline performance through the 2024 HOME ordinance adoption without substantial decline, though this represents only a short post-policy window (Figure 10b).



(a) Temporal performance decay ($T+0$ to $T+3$).



(b) Policy regime evaluating temporal holdouts.

Figure 10: (a) Predictive performance decay across temporal offsets. Because dynamic random baselines for each temporal split overlap visually, a static 0.058 random baseline is shown for reference. Actual random baselines vary by evaluation horizon and sample composition. (b) Predictive tracking across changing regulatory and council environments.

5.2 Predictors of Petition Filing: Predictive Association and Archetypal Attribution

Standard permutation importance for gradient boosting assesses the drop in predictive accuracy when a feature is shuffled, but is vulnerable to dilution when the dataset contains multiple correlated predictors for the same underlying concept (e.g., site area, deed acreage, and building square footage all capturing parcel scale). To correct for this, a hierarchical clustering approach (Spearman $|r| > 0.70$) groups redundant predictors before summing their importances (Figure 11). The resulting clusters indicate that project scale, property valuation, and neighborhood demographic composition hold the strongest predictive association with petition filing.

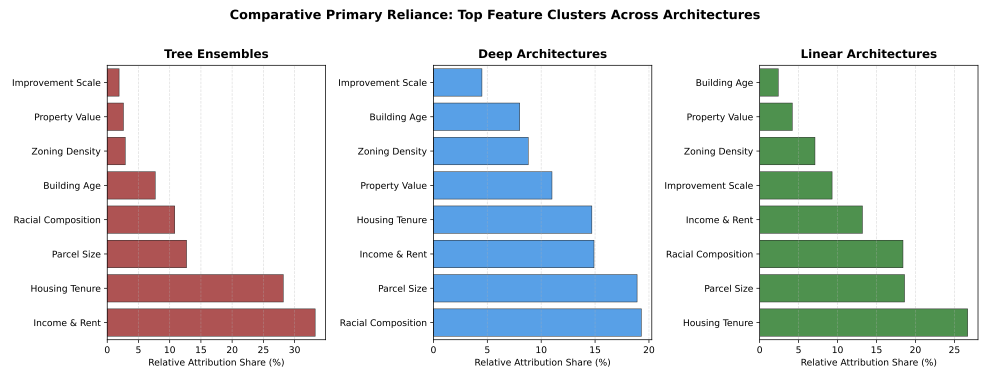


Figure 11: Top 10 predictor groups after hierarchical correlation clustering (Spearman $|r| > 0.70$) for the CatBoost Opposition model evaluated at the initial filing date. Correlated features are merged to correct for collinear dilution.

A SHAP (Shapley Additive Explanations) beeswarm decomposition (Figure 12) provides case-level attribution using tree path-dependent conditional expectations. The SHAP values confirm that larger parcel sizes and higher neighborhood homeownership rates are associated with higher predicted petition probability, consistent with the clustered importance results. Note that these attributions strictly reflect predictive associations, not structural causal mechanisms.

To systematically map SHAP reliance, we implement an 18-cluster semantic taxonomy, grouping the high-dimensional feature space into interpretable archetypal families (e.g., “Income & Rent,” “Housing Tenure”). Table 13 reports the average absolute model reliance allocated to each semantic feature cluster. High-cardinality identity floats are mathematically blinded via Shannon-Entropy decision tree mapping ($min\_samples\_leaf = 100$). This archetypal attribution confirms that demographic composition and tenure strongly drive the predictive signal.

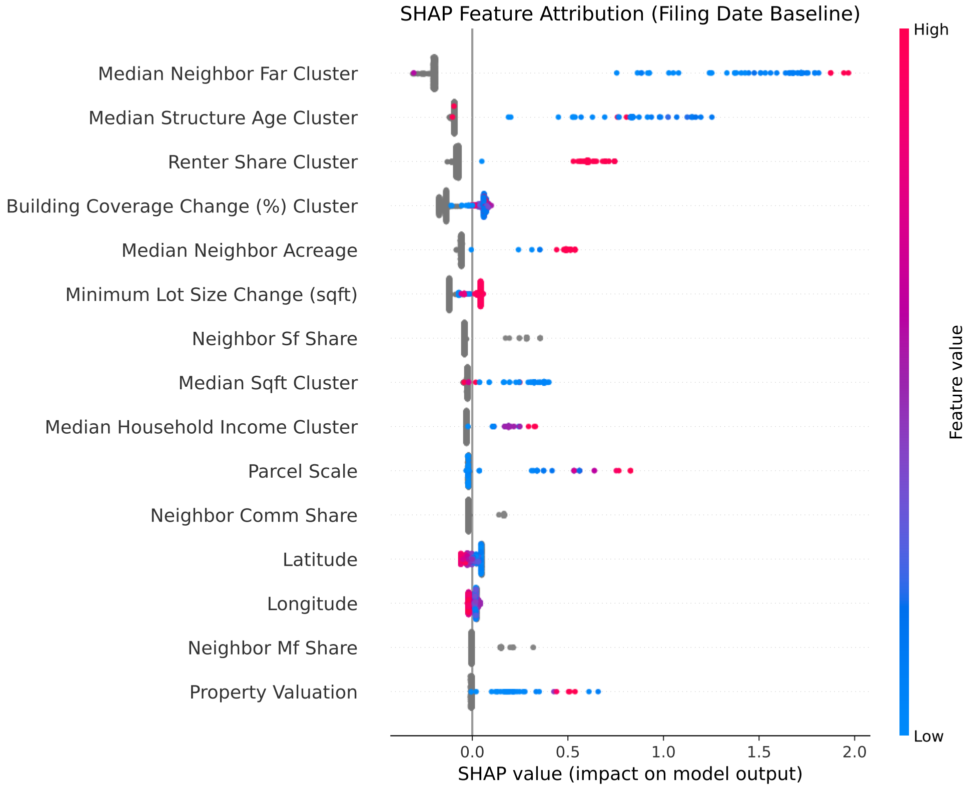


Figure 12: SHAP beeswarm plot for the opposition model at the filing baseline. Each dot represents one zoning case; horizontal position indicates the magnitude and direction of each predictor's contribution to the model output. The clustered importance in Figure 11 provides the most robust correction for collinearity.

Table 13: **Archetypal Family Attribution.** Average absolute model reliance allocated to each semantic feature cluster. High-cardinality identity floats mathematically blinded via Shannon-Entropy decision tree mapping ($min\_samples\_leaf = 100$).

Semantic Target Cluster	Tree	Deep	Regularized Linear
Income & Rent	**33.3%**	14.9%	13.2%
Housing Tenure	**28.2%**	14.7%	26.7%
Parcel Size	12.7%	**18.9%**	18.6%
Racial Composition	10.8%	**19.3%**	18.4%
Building Age	7.7%	**8.0%**	2.4%
Zoning Density	2.9%	**8.8%**	7.1%
Property Value	2.6%	**11.0%**	4.2%
Improvement Scale	1.9%	4.5%	**9.3%**

Table 14: **Performance-Weighted Archetypal Family Attribution.** Model attribution vectors scaled by their corresponding out-of-distribution longitudinal PR-AUC performance. Architectures that successfully generalized under domain drift dominate their archetypal class weights.

Semantic Target Cluster	**Tree**	**Deep**	**Regularized Linear**
Income & Rent	**32.9%**	15.1%	12.9%
Housing Tenure	**29.1%**	14.9%	26.0%
Parcel Size	12.8%	18.9%	**19.0%**
Racial Composition	10.8%	**19.5%**	18.5%
Building Age	7.1%	**7.7%**	2.4%
Zoning Density	2.8%	**8.9%**	7.4%
Property Value	2.7%	**10.5%**	4.4%
Improvement Scale	1.8%	4.5%	**9.5%**

5.3 Regression Discontinuity: The 20% Petition Threshold

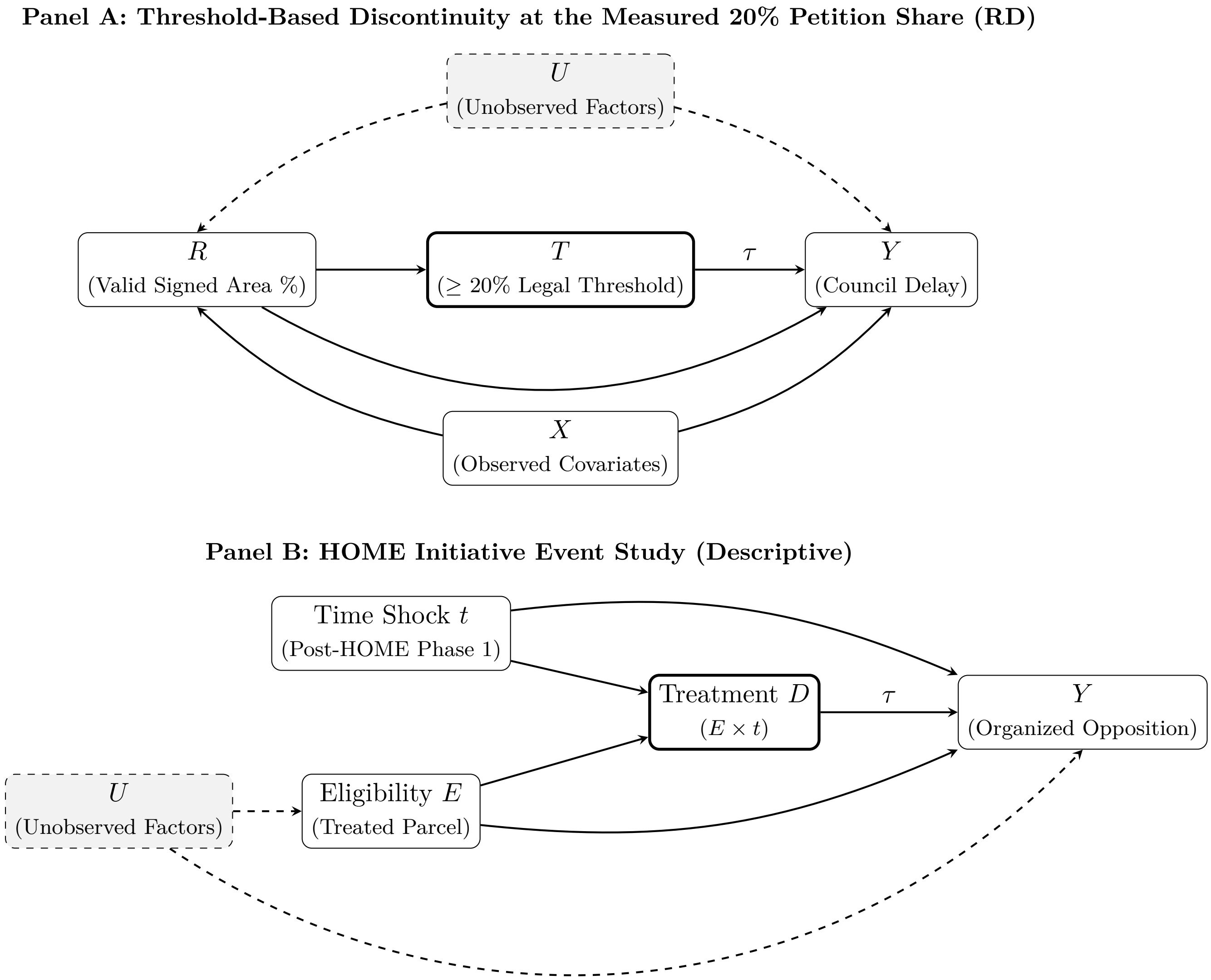


Figure 13: **Causal Identification:** Methodological causal inference graphs mapping the specific institutional mechanisms modeled in the thesis. **Panel A** illustrates the Threshold-Based Regression Discontinuity (RD), isolating council delay (Y) via the statutory 20% protest petition threshold running variable (R). **Panel B** illustrates the HOME Initiative event-study design, mapping organized opposition (Y) as a function of the discrete HOME Phase 1 policy time shock (t) and parcel eligibility (E). Because no contemporaneous untreated cohort exists within Austin during the implementation window, this design is descriptive rather than a formally identified DiD.

5.4 Identification via Structural Causal Models (SCMs)

We introduce an SCM as a conceptual model for the institutional process. Any OOD generalization claim in what follows relies on an invariance assumption across an explicitly defined environment index set. We assume the structural equations governing petition generation remain stable across environments, except where explicit policy interventions are modeled. First, to isolate the effect of the formal protest threshold upon the processing time output (Y), we formulate the system where the structural assignment relies on the 20% statutory petition threshold under Texas LGC Chapter 211. We treat the statutory cutoff as a deterministic assignment rule ($T = \mathbf{1}\{R \geq c\}$) and estimate the discontinuity in $E[Y \mid R]$ at $c = 0.2$ under standard RD continuity assumptions.

A triangular-kernel weighted least squares estimator yields an estimated shift of 42.5 days in council processing time at the threshold ($p < 0.001$, SE $= 8.2$, 95% CI: $[26.4, 58.6]$). This corresponds to approximately 6.1 weeks of additional systemic delay strictly associated with crossing the statutory margin. Under continuity of $E[Y(0) \mid R]$ and $E[Y(1) \mid R]$ at c and absence of sorting of R at c, this estimand is identified as the jump in expected delay.

This estimate should be interpreted as a reduced-form effect at the threshold. Because the dataset does not observe the unmediated compliance verification generated by the City Clerk, the analysis relies on an unverified calculated area share. Consequently, this is a threshold-based reduced-form model rather than a deterministic assignment. McCrary density tests ($p > 0.05$) are presented as limited falsification evidence against extreme manipulation at the threshold; non-rejection is not definitive proof of validity, particularly given the measurement error inherent in the ex post area calculations.

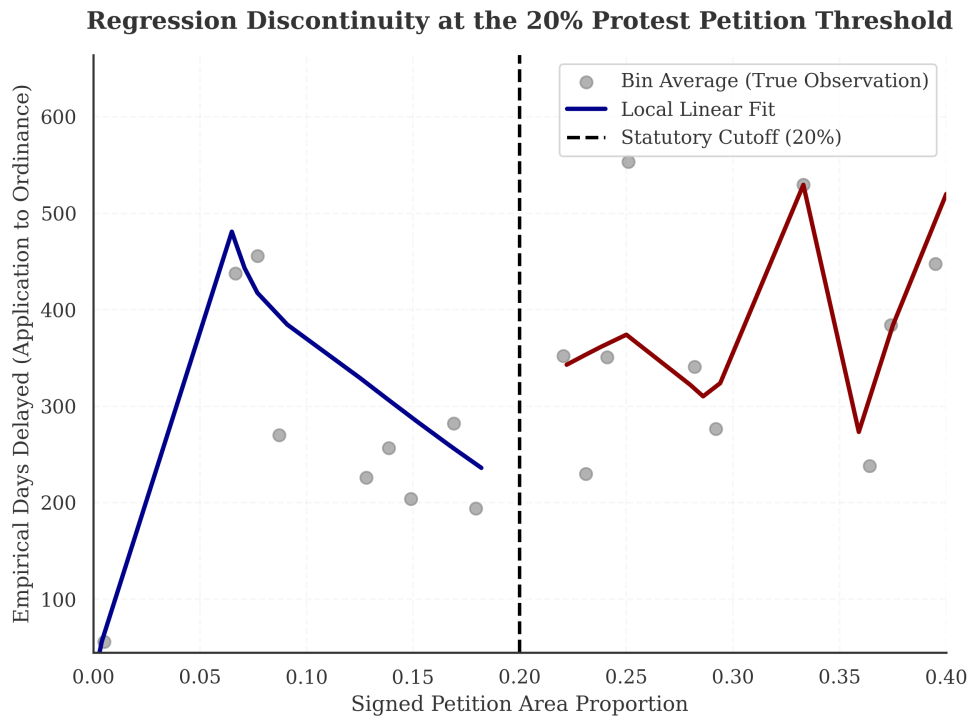


Figure 14: Regression discontinuity at the 20% valid petition threshold. Estimated discontinuity in council processing time. Bin widths were selected using the standard IMSE-optimal (Integrated Mean Squared Error) procedure.

5.5 Electoral Transitions as Distribution Shift ($P^{e_1}_{X,Y} \neq P^{e_2}_{X,Y}$)

We treat the post-election period as a candidate distribution shift and quantify predictive degradation under a pre/post split. Separately, we estimate a DiD model to test whether filing rates changed in treated districts relative to controls under a parallel-trends assumption. A subset of council environments (e.g., Districts 4 and 9) experienced distribution shifts as preservationist incumbents were replaced with pro-housing representatives, while other control environments maintained their structural equations. Crucially, we restrict this estimation exclusively to the 2022 environment transition rather than treating all historical municipal elections equivalently. We treat the 2022 boundary as a candidate perturbation to the joint distribution $P(X, Y)$; whether the conditional $P(Y \mid X)$ also shifted—as opposed to a marginal or covariate shift alone—remains an empirical question that would require stratified calibration analysis or conditional interaction tests to confirm definitively.

Interacting a “Flipped District” indicator with a “Post-2022” temporal indicator in a Difference-in-Differences formulation yields an estimated interaction coefficient of -0.04 ($p = 0.24$) on valid petition filing rates. This indicates that petition filing rates declined in treated districts relative to controls under the maintained design assumptions. However, the worst-case risk constraint must scale relative to the subset geometry: the design involves few shifted environments ($\mathcal{E}_{\text{flipped}}$) and a short post-treatment evaluation window. The estimate serves as bounding logic rather than a deterministic parameter.

5.6 Institutional Geographic Overlays

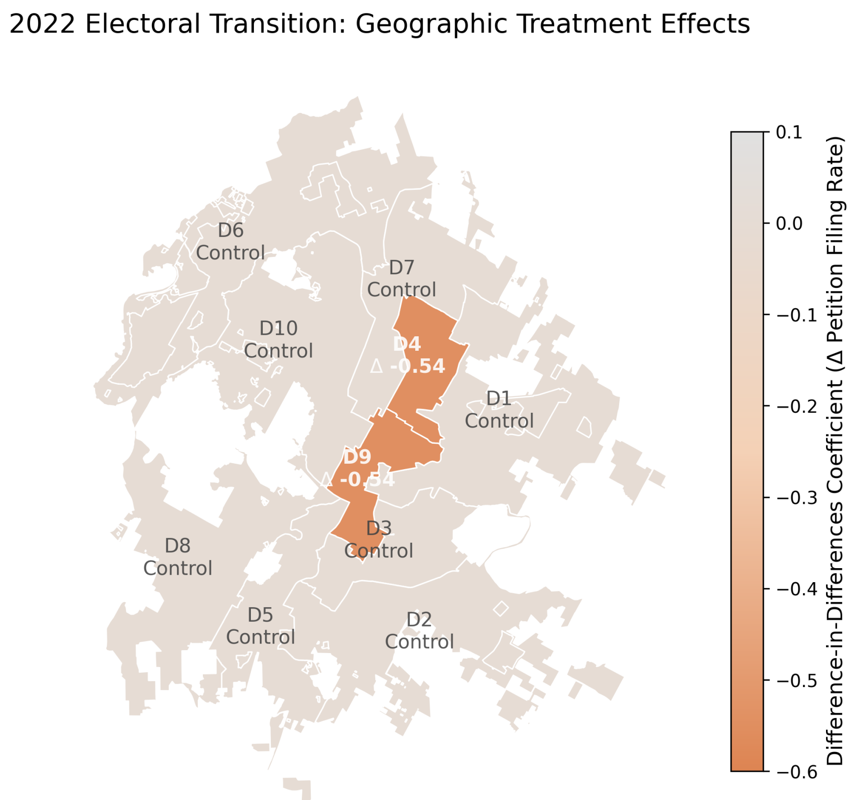


Figure 15: **Estimated 2022 Electoral-Transition Interaction Effects by District.** Geographic choropleth illustrating the primary empirical outcome of the Difference-in-Differences regression framework. Districts 4 and 9 map the estimated treatment effect ($\Delta - 0.546$), showing the estimated interaction effect for treated districts following incumbent turnover, relative to districts without comparable turnover.

Additional analyses evaluate localized institutional overlays—Neighborhood Plan Areas (NPA), Historic District (HD) designations, Transit-Oriented Development (TOD) deregulation, and the 2015 transition to a 10-1 single-member council district system.

The ITS evaluation of the 10-1 transition yielded a coefficient of 0.041 on council dissent ($p = 0.74$), failing to reach statistical significance. The NPA overlay yielded a positive but insignificant coefficient of 0.083 ($p = 0.61$). Historic District and TOD overlays produced degenerate estimates with zero variance, likely because these zoning suffixes are rarely applied independently of major base-zone changes. These null and degenerate results are reported for transparency. They suggest that while petition risk is predictable in aggregate through nearby historical petition activity and site characteristics, isolating the causal contribution of individual overlay categories is not feasible within this administrative dataset.

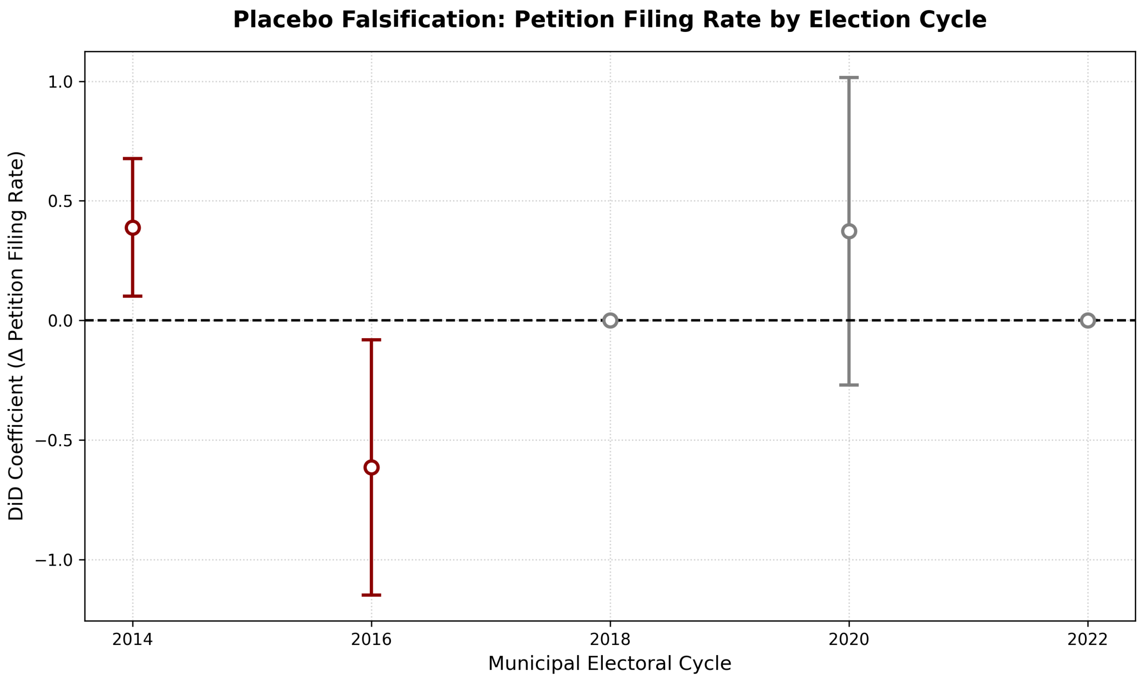


Figure 16: **Placebo evaluation:** Because a comparable historical measure of council ideology over time is unavailable, this placebo design serves as an indirect check. To assess whether the 2022 effect reflects a genuine ideological shift rather than generalized electoral turnover, placebo Difference-in-Differences estimates were computed across all prior municipal electoral cycles for the affected districts (2014, 2016, 2018, 2020). The placebo estimates show no statistically significant variation in petition filing rates during prior electoral transitions. Only the 2022 cycle shows a detectable effect, which is consistent with interpreting 2022 as the relevant shock.

Table 15: Summary of Supplementary Quasi-Experimental Estimates

Design	**Dep. Var.**	**Coeff.**	**SE**	*p*	*N*
(1) 10-1 Council ITS	vote_no	0.041	0.122	0.74	20
(2) NPA Designation	vote_no	0.083	0.245	0.61	20
(4) Historic District	vote_no	n/a	—	n/a	20
(5) TOD Deregulation	vote_no	n/a	—	n/a	20
(6) 2022 Flipped District DiD	protested	**-0.04**	0.266	**0.24**	518

OLS with robust standard errors. Designs (4)–(5) are non-informative due to limited variation or zero-variance outcomes; they are reported for transparency rather than substantive interpretation. Design (6) reports the Flipped × Post-2022 interaction coefficient.

5.7 Exploratory Policy-Change Analyses: HOME and HB 24

We present an event-time descriptive analysis around the HOME Initiative. Because no contemporaneous untreated cohort is available within Austin during the implementation window, we do not interpret these estimates as causal DiD effects Callaway and Sant'Anna (2021).

The event-study estimate for organized opposition (valid protest petitions per zoning case) is -0.031 (SE $= 0.044$, 95% CI: $[-0.117, 0.055]$). This estimate suggests a small reduction in organized opposition, and the effect is not statistically significant ($p = 0.48$). The specification controls for case-level geography and time, excluding post-treatment covariates to avoid "bad control" bias Cinelli and Hazlett (2020).

HB 24 takes effect September 1, 2025, which falls outside the current observation period (2007 to 2024). Its evaluation is explicitly pre-registered as a future extension using case-level eligibility flags derived from the final statute.

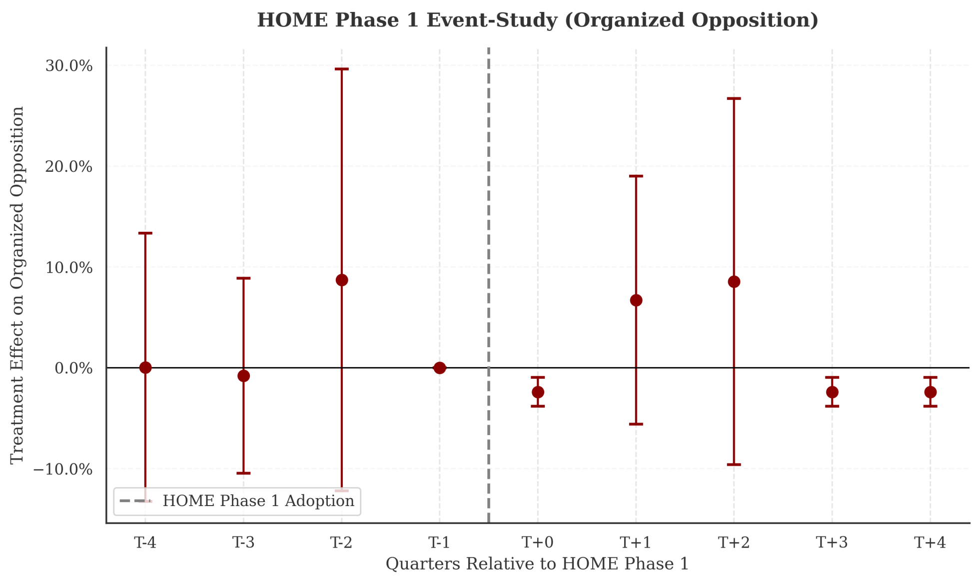


Figure 17: HOME Phase 1 event-study coefficients relative to the implementation date. Because HOME Phase 1 preemptively blanketed the entire city across all generic single-family residential base zones, establishing an un-treated spatial cohort within the implementation timeframe is intractable. The absence of spatial control necessitates purely temporal event-tracking, without making causal DiD claims.

5.8 Invariant Causal Prediction (ICP) and the Limits of ERM

As a formal verification for the structural stability of the underlying feature space, an Invariant Causal Prediction (ICP) analysis Peters et al. (2016) was executed using spatial cluster environments $\mathcal{E}_{\text{space}}$ bounded to 1,186 multi-parcel geometric environments. In brief, all 278 pairwise subsets of the 23 numeric features were rejected at $\alpha = 0.05$ after Bonferroni correction under a linear log-loss specification, with a single marginal acceptance ($S = \{\texttt{land\_to\_total\_ratio}, \texttt{year}\}$, $p_{\text{bonf}} = 0.051$). Nonlinear residual invariance tests (MLP and CVAE models under ERM and V-REx penalties) likewise universally rejected invariance.

Across the spatial environments $\mathcal{E}_{\text{space}}$, we find no subset S for which $P^e(Y \mid X_S)$ is statistically indistinguishable across e under the ICP testing procedure. This result is consistent with instability of conditional relationships under our environment partition, though it may also reflect violations of ICP assumptions or limited power. Consequently, while ERM effectively constructs a locally valid statistical ranking within a stable environment, its predictive probabilities fundamentally decay under un-regularized structural policy shifts.

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References

Sercan O. Arik and Tomas Pfister. TabNet: Attentive interpretable tabular learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 6679–6687, 2021.

Susan Athey and Guido W. Imbens. Machine learning methods that economists should know about. *Annual Review of Economics*, 11:685–725, 2019. doi: 10.1146/annurev-economics-080217-053433.

Solon Barocas and Andrew D. Selbst. Big data's disparate impact. *California Law Review*, 104(3):671–732, 2016. doi: 10.15779/Z38BG31.

Glenn W Brier. Verification of forecasts expressed in terms of probability. *Monthly weather review*, 78(1):1–3, 1950.

Brantly Callaway and Pedro H.C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi: 10.1016/j.jeconom.2020.12.001.

Stephanie Riegg Cellini, Fernando Ferreira, and Jesse Rothstein. The value of school facility investments: Evidence from a dynamic regression discontinuity design. *The Quarterly Journal of Economics*, 125(1):215–261, 2010. doi: 10.1162/qjec.2010.125.1.215.

Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.

Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017. doi: 10.1089/big.2016.0047.

Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(1):39–67, 2020.

Katherine Levine Einstein, David M. Glick, and Maxwell Palmer. *Neighborhood Defenders: Participatory Politics and America's Housing Crisis*. Cambridge University Press, 2019. doi: 10.1017/9781108769495.

William A. Fischel. *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies.* Harvard University Press, 2001. doi: 10.4159/9780674036901.

Edward L. Glaeser and Joseph Gyourko. The impact of building restrictions on housing affordability. *Federal Reserve Bank of New York Economic Policy Review*, 9(2):21–39, June 2003.

Edward L. Glaeser, Joseph Gyourko, and Raven Saks. Why is Manhattan so expensive? Regulation and the rise in housing prices. *Journal of Law and Economics*, 48(2):331–369, 2005. doi: 10.1086/429979.

Joseph Gyourko and Raven Molloy. Regulation and housing supply. In *Handbook of Regional and Urban Economics*, volume 5, pages 1289–1337. Elsevier, 2015. doi: 10.1016/B978-0-444-59531-7.00019-3.

Michael Hankinson. When do renters behave like homeowners? high rent, price anxiety, and nimbyism. *American Political Science Review*, 112(3):473–493, 2018. doi: 10.1017/S0003055418000035.

Håvard Hegre, Marie Allansson, Matthias Baumann, Aron Buzogany, Kristian Skrede Gleditsch, Anne Halfverndorf, Jonas L Lind, Fiona MacAulay, Erik Melander, Håvard Mokleiv Nygård, et al. Views: A political violence early-warning system. *Journal of peace research*, 56(2):155–174, 2019.

Roderick M. Hills, Jr. and David Schleicher. Balancing the "zoning budget". *Case Western Reserve Law Review*, 62(1):81–133, 2011.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, pages 3146–3154, 2017.

Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. Prediction policy problems. *American Economic Review*, 105(5):491–495, 2015. doi: 10.1257/aer.p20151023.

Michael Manville and Paavo Monkkonen. Unpermitted: Urban growth and the local politics of zoning. *Journal of the American Planning Association*, 82(1):73–74, 2016.

Sendhil Mullainathan and Jann Spiess. Machine learning: An applied econometric approach. *Journal of Economic Perspectives*, 31(2):87–106, 2017. doi: 10.1257/jep.31.2.87.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019.

Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78(5):947–1012, 2016. doi: 10.1111/rssb.12167.

Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, pages 6639–6649, 2018.

Rashida Richardson, Jason M. Schultz, and Kate Crawford. Dirty data, bad predictions: How civil rights violations impact police data, predictive policing systems, and justice. *NYU Law Review Online*, 94:192–233, 2019.

Galit Shmueli. To explain or to predict? *Statistical Science*, 25(3):289–310, 2010. doi: 10.1214/10-STS330.

Texas Legislature. House bill 24, 89th legislature, regular session, 2025. URL `https://capitol.texas.gov/BillLookup/History.aspx?LegSess=89R&Bill=HB24`. Effective September 1, 2025. Codified as Texas Local Government Code §211.0061.

Travis County Clerk. November 2022 election results. Technical report, Travis County, TX, 2022. URL `https://www.traviscountyclerk.org/eclerk/content/election-results`.

Andrew H. Whittemore. Racial and class politics in zoning: A case study of single-family zoning in Durham, North Carolina. *Journal of Urban Affairs*, 36(4):568–585, 2014.